

Warehouse & Retail Sales Analysis

SQL Server + Power BI Dashboard Project

Operational volume analysis by category, supplier, product, and reporting period

Project Overview

This project analyzes warehouse and retail sales data to understand product movement across different channels, categories, suppliers, and time periods. The goal was to clean, validate, analyze, and visualize operational sales volume data using Excel Power Query, SQL Server, and Power BI.

The analysis focuses on movement volume, not revenue, because the dataset does not include price, cost, or profit information.

Tools Used

- Excel / Power Query - data cleaning and preparation
- SQL Server - data storage, validation, and analysis
- SQL Server Management Studio - SQL queries and data exploration
- Power BI - dashboard creation and visualization

Dataset

The dataset contains warehouse and retail sales records with the following main fields:

Field	Meaning
YEAR	Year of the reporting period
MONTH	Month of the reporting period
SUPPLIER	Supplier or distributor name
ITEM CODE	Product identifier
ITEM DESCRIPTION	Product name or description
ITEM TYPE	Product category/type
RETAIL SALES	Volume sold through retail channel
RETAIL TRANSFERS	Volume transferred to retail channel
WAREHOUSE SALES	Volume moved or sold through warehouse channel
DATE	Combined date field created from YEAR and MONTH

Final cleaned dataset summary:

Metric	Value
Rows	307,645
Unique products	34,056
Suppliers	397
Item types	9
Reporting periods	24
Date range	June 2017 to September 2020

Data Cleaning Process

- Promoted headers
- Corrected data types
- Converted product codes to text to preserve identifier formatting
- Replaced missing supplier values with Unknown Supplier
- Replaced missing item type values with Unknown Item Type
- Converted sales and transfer columns to decimal numbers
- Created a proper DATE field from YEAR and MONTH
- Verified that the dataset contained 307,645 complete rows after cleaning

SQL Validation

After cleaning the data, it was imported into SQL Server for validation and analysis.

```
SELECT COUNT(*) AS total_rows
FROM [WarehouseRetail].[dbo].[Warehouse_and_Retail_Sales$];
```

Result: 307,645 rows

- Minimum and maximum dates
- Number of reporting periods
- Number of unique suppliers
- Number of unique products
- Number of item types
- Data types after import

Key Metrics

- Total Warehouse Sales
- Total Retail Sales
- Total Retail Transfers
- Unique Products
- Unique Suppliers
- Unique Item Types
- Total Movement

Total Movement was calculated as:

```
Total Movement =  
SUM('Warehouse_and_Retail_Sales$'[RETAIL SALES]) +  
SUM('Warehouse_and_Retail_Sales$'[RETAIL TRANSFERS]) +  
SUM('Warehouse_and_Retail_Sales$'[WAREHOUSE SALES])
```

Dashboard

- KPI cards for total movement metrics
- Monthly sales and transfers trend
- Warehouse sales by item type
- Retail sales by item type
- Top suppliers by warehouse sales
- Top 10 products by total movement
- Interactive filter by item type

Key Insights

1. Warehouse is the main driver

The majority of all operational volume comes from Warehouse Sales, totaling around 7.78 million units. This shows that the warehouse is the primary channel driving the business's operations.

2. Beer dominates the warehouse

Inside the warehouse, Beer is the clear leader. Its volume is significantly higher than wine or any other category, meaning warehouse activity relies heavily on this single product type.

3. Retail sales are more balanced

Unlike the warehouse, Retail Sales are evenly spread out across liquor, wine, and beer. Customer buying habits in stores are more diverse rather than being focused on just one category.

4. Top products are well-known Beer SKUs

The top 10 most-moved products are almost exclusively Beer brands (such as Corona, Heineken, Miller Lite, Modelo, and Bud Light). This indicates that a small, specific group of popular items drives a massive share of the workload.

5. Negative values represent operational adjustments

Some categories (like REF and DUNNAGE) show negative numbers. These were kept in the data because they likely represent returns, credits, or logistics corrections. They should be analyzed on their own, separate from standard sales.

Business Interpretation

The analysis shows that warehouse operations are heavily driven by Beer products, while retail activity is more distributed across Liquor, Wine, and Beer. From an operations perspective, Beer products may require stronger inventory planning, supplier coordination, and warehouse capacity management due to their high movement volume.

Categories with negative values should be reviewed separately to understand whether they represent returns, credits, or inventory adjustments.

Limitations

- The dataset does not include price, cost, revenue, or profit.
- Sales values were treated as volume metrics, not monetary values.
- The dataset does not contain every month consecutively.
- External factors such as promotions, seasonality, demand changes, or market events were not included.
- Negative values require business context to determine their exact meaning.

Conclusion

This project demonstrates a complete basic data analysis workflow:

1. Data cleaning in Power Query
2. Data validation in SQL Server
3. Exploratory analysis with SQL
4. Dashboard creation in Power BI
5. Business insight generation from operational data

The final dashboard provides a clear overview of warehouse and retail product movement, category performance, supplier contribution, product concentration, and monthly trends.

Recommended repository title: Warehouse & Retail Sales Analysis | SQL + Power BI Dashboard